

From Static Pre-training to Dynamic Adaptation: Navigating the Stability-Plasticity Dilemma in Foundation Models

The emergence of Foundation Models (FMs) has precipitated a paradigm shift from static deployment toward dynamic adaptation, necessitating robust Continual Learning (CL) mechanisms to counter knowledge staleness [6]. However, naive fine-tuning often induces catastrophic forgetting, where adaptation to new tasks severely degrades prior capabilities, highlighting a critical stability-plasticity dilemma [33]. While Parameter-Efficient Fine-Tuning (PEFT) methods like Low-Rank Adaptation offer computational efficiency, empirical studies reveal they remain susceptible to forgetting and "catastrophic inheritance" of biases if not geometrically constrained [24] [75]. Consequently, research is increasingly pivoting toward geometry-aware strategies, such as orthogonal gradient projections and feature-space alignment, to mathematically guarantee the preservation of pre-trained representations [298] [85]. Furthermore, the field is re-evaluating the locus of adaptation, with evidence suggesting that Reinforcement Fine-Tuning and modular memory architectures may inherently mitigate forgetting better than traditional supervised approaches [122] [182]. This review synthesizes these divergent pathways, arguing that sustainable FM evolution requires moving beyond simple weight updates to unified frameworks integrating geometric constraints, selective unlearning, and compositional ecosystems [144].

Introduction: The Paradigm Shift from Static Pre-training to Dynamic Adaptation

The emergence of Foundation Models (FMs) has fundamentally revolutionized artificial intelligence, establishing new benchmarks in natural language processing, computer vision, and specialized domains through massive-scale pre-training on diverse datasets. These models, characterized by their enormous parameter counts and exposure to internet-scale knowledge, possess remarkable zero-shot generalization capabilities and rich semantic representations. However, the prevailing paradigm of static pre-training renders these systems inherently susceptible to knowledge staleness and distribution shifts as the real world evolves. While centralized, monolithic models can tackle diverse tasks, their inability to continuously acquire, retain, and refine knowledge over time poses a critical limitation for deployment in dynamic environments. Consequently, the research community is witnessing a significant paradigm shift from static deployment toward dynamic adaptation, where Continual Learning (CL) is no longer viewed as obsolete but as an essential mechanism for ensuring FMs remain up-to-date, specialized, and modular [6].

A primary driver for this shift is the phenomenon of catastrophic forgetting, where fine-tuning a model on new tasks or data streams leads to a severe degradation of performance on previously learned concepts. This stability-plasticity dilemma is particularly acute in FMs; while fine-tuning improves adaptation to new domains, it often cripples the model's ability to recognize concepts unrelated to the downstream task, effectively erasing the vast knowledge acquired during pre-training [33]. Empirical studies across various modalities confirm this vulnerability. In the domain of time series, Foundation Models (TSFMs) exhibit significant forgetting when fine-tuned sequentially on multiple datasets, illustrating a fundamental trade-off between adapting to new periodic structures and retaining prior forecasting capabilities [5]. Similarly, in medical imaging, the scarcity of data necessitates sequential adaptation across different modalities and anatomical regions, yet standard fine-tuning approaches struggle to maintain integrity across these diverse tasks without explicit continual learning frameworks [49]. The issue extends beyond mere performance drops; it involves the "catastrophic inheritance" of biases and limitations from pre-training data, which can be exacerbated or altered during downstream adaptation, leading to security vulnerabilities and value misalignment if not properly managed [75].

To address these challenges, the field is moving beyond traditional end-to-end fine-tuning toward more sophisticated adaptation strategies that balance efficiency with knowledge preservation. One prominent direction involves Parameter-Efficient Fine-Tuning (PEFT) and selective parameter updates. Methods such as Low-Rank Adaptation (LoRA) and sparse parameter updates allow models to localize changes to specific subsets of parameters relevant to new tasks, thereby preserving the generalizability of the foundation model while achieving high accuracy on downstream objectives [18]. Research indicates that the rank of adapters and the specific layers updated play a crucial role in mitigating forgetting, with some studies suggesting that vision transformers exhibit unique "contextual" forgetting behaviors distinct from residual networks when subjected to incremental low-rank updates [24]. Furthermore, novel regularization techniques like Proxy-FDA have been developed to align feature distributions structurally rather than point-wise, significantly reducing concept forgetting by preserving the local neighborhood structures of pre-trained features [85].

Another critical aspect of this paradigm shift is the exploration of alternative learning mechanisms beyond weight updates. The integration of In-Context Learning (ICL) with In-Weight Learning (IWL) through modular memory architectures offers a promising path forward, allowing agents to accumulate experience rapidly without destabilizing the core model parameters [182]. Additionally, the role of the learning paradigm itself is being re-evaluated; recent findings suggest that Reinforcement Fine-Tuning (RFT) inherently mitigates forgetting better than Supervised Fine-Tuning (SFT) due to implicit regularization effects, challenging the assumption that SFT is the default standard for post-training [122]. Theoretical analyses further reveal that forgetting is often driven by gradient conflicts and geometric misalignments between tasks, leading to the development of geometry-aware merging and update strategies that gate integration based on task compatibility [261].

Despite these advancements, significant gaps remain in understanding the long-term sustainability and scalability of continual adaptation. The environmental cost of continual training is an underexplored dimension, with evidence suggesting that inference energy consumption and the complexity of CL algorithms vary drastically, necessitating a

"Green AI" perspective on model evolution [46]. Moreover, the concept of unlearning is gaining traction as a counterpart to continual learning, where the goal shifts from merely adding knowledge to selectively removing specific data influences or capabilities, a requirement driven by privacy regulations and the need for model corrigibility [59]. As the field progresses, the focus is increasingly turning toward compositional approaches and ecosystems of interacting models rather than single static entities, marking a rebirth of continual learning as the cornerstone of next-generation intelligent systems [144].

The Stability-Plasticity Dilemma and the Distinction Between CL and Editing

The transition from static pre-training to dynamic adaptation in foundation models is fundamentally constrained by the stability-plasticity dilemma, where the capacity to acquire new information often compromises the retention of previously learned knowledge. While foundation models exhibit robust zero-shot generalization across diverse domains, naive fine-tuning frequently induces catastrophic forgetting, degrading performance on prior tasks and eroding the broad conceptual understanding gained during pre-training [5]. This phenomenon is not merely a loss of accuracy but represents a structural drift in the model's functional directions, where updates for new tasks interfere with the geometric subspaces encoding historical knowledge [298]. Theoretical analyses suggest that this forgetting is driven by strongly negative gradient similarities between old and new tasks, indicating that standard optimization trajectories inherently conflict with the preservation of established representations [117]. Furthermore, fine-tuning can introduce "intruder dimensions" into the representation space, overspecializing the model to limited downstream data and destroying the rich feature neighborhoods that encode general concepts [135].

A critical distinction emerges between Continual Learning (CL) and Model Editing, though both aim to update model behavior without full retraining. CL typically addresses sequential task acquisition, striving to maintain performance across a growing distribution of tasks, whereas model editing often targets precise, localized updates to specific facts or behaviors [17]. However, the boundary blurs as both fields grapple with the same underlying mechanism: the overwrite of parameter space. In CL, methods like Dynamic Orthogonal Continual (DOC) fine-tuning attempt to mitigate interference by adjusting gradients to be orthogonal to tracked historical function directions, effectively creating disjoint update paths for sequential tasks [298]. Similarly, in the realm of editing, approaches such as X-Edit utilize null-space projections to ensure that parameter updates for correcting specific errors remain orthogonal to the representations of anchor samples, thereby providing mathematical guarantees against perturbing established diagnostic capabilities [219]. Despite these geometric parallels, the objectives diverge; CL seeks broad retention across task distributions, while editing prioritizes the surgical modification of specific knowledge units, often risking "generic knowledge loss" if the sparsity of updates is not carefully managed [18].

The degradation of zero-shot capabilities following fine-tuning highlights the fragility of pre-trained features when subjected to end-to-end optimization. To counter this, recent methodologies have shifted from weight-space constraints to feature-space regularization. For instance, Proxy-FDA aligns the local structures of feature distributions using synthetic proxies, preserving the semantic relationships inherent in the pre-trained model rather than just matching point-wise features [85]. Furthermore, the distinction between learning paradigms plays a significant role; Reinforcement Fine-Tuning (RFT) has been shown to inherently preserve prior knowledge better than Supervised Fine-Tuning (SFT) due to the implicit regularization provided by on-policy data, suggesting that the learning objective itself dictates the severity of the stability-plasticity trade-off [122]. This implicit regularization allows RFT to protect and even enhance general capabilities, whereas SFT often leads to severe degradation of the model's foundational skills [143].

Contradictions and nuances exist regarding the efficacy of parameter-efficient methods versus full fine-tuning. While Low-Rank Adaptation (LoRA) is widely adopted for its efficiency, empirical evidence suggests that merging LoRA adapters sequentially can still lead to significant forgetting, particularly in vision transformers which exhibit "contextual" forgetting behaviors not seen in residual networks [24]. Conversely, some approaches argue that freezing the backbone entirely and relying on task-specific adapters offers a practical solution for few-shot continual learning, effectively eliminating forgetting by design, albeit potentially at the cost of optimal plasticity for complex task shifts [34]. Additionally, the concept of "Catastrophic Inheritance" posits that foundation models may inherit biases and limitations from pre-training data that are exacerbated rather than corrected by naive adaptation, complicating the distinction between forgetting old knowledge and failing to overcome inherited deficiencies [75]. Advanced strategies now explore the interplay between memory mechanisms and adaptation, proposing hybrid solutions where rapid adaptation occurs via context while stable knowledge is consolidated in long-term memory modules [182]. Moreover, post-hoc repair methods like Spectral Unforgetting demonstrate that a portion of capability loss is attributable to removable spectral noise in the weight update rather than an unavoidable consequence of specialization, offering a pathway to recover damaged capabilities without retraining [205]. Ultimately, achieving the upper bound of joint

training accuracy in continual settings requires moving beyond simple regularization to principled approaches that leverage task geometry and causal inference to distinguish between transferable and conflicting knowledge [250].

Taxonomy of Adaptation Strategies and Evaluation Challenges

The paradigm shift from static pre-training to dynamic adaptation has necessitated a rigorous taxonomy of strategies designed to update foundation models in response to evolving data distributions. Current literature broadly categorizes these approaches into offline and online settings, distinguished by the availability of historical data and the immediacy of the adaptation requirement. In offline settings, where batch processing of historical data is feasible, strategies often revolve around continual pre-training and instruction tuning to mitigate knowledge staleness. For instance, researchers have proposed dynamically adjusting data proportions based on learning velocity to optimize domain-adaptive pre-training, favoring slower-learning domains to ensure comprehensive coverage [289]. Similarly, addressing redundancy in visual instruction datasets, other works utilize adaptive data selection to mitigate catastrophic forgetting during multimodal instruction tuning by balancing sample efficiency and effectiveness [ADAPT-\$\infty\$: SCALABLE CONTINUAL MULTIMODAL INSTRUCTION TUNING VIA DYNAMIC DATA SELECTION]. Conversely, online settings demand immediate responses to streaming data, often without access to past examples, requiring algorithms that can detect task switches and distribution shifts in non-stationary environments [194]. To handle architectural heterogeneity in real-time user response systems, frameworks have been presented that combine offline embedding transfer with online asymmetric co-distillation, allowing students to update frequently while teachers update infrequently [156].

A critical divergence in adaptation strategies lies in the locus of modification: parameter space versus activation space. Traditional approaches predominantly rely on weight updates, utilizing techniques such as Low-Rank Adaptation (LoRA) or full fine-tuning, yet these methods frequently induce catastrophic forgetting where new knowledge overwrites prior capabilities. To counter this, studies demonstrate that reinforcement learning paradigms inherently preserve prior knowledge better than supervised fine-tuning due to their mode-seeking nature and use of on-policy data [143]. Additional research confirms that reinforcement fine-tuning naturally mitigates forgetting in continual post-training scenarios, protecting and even enhancing general model capabilities [122]. Furthermore, attributing forgetting to misalignments in the covariance geometry of task updates, some scholars propose geometry-aware merging to control integration and prevent state-relative update-integration failures [261]. In contrast to weight-based modifications, emerging research advocates for steering internal activations as a distinct adaptation paradigm enabling local and reversible behavioral changes without parameter updates [108]. This perspective is corroborated by automated, lightweight approaches for constructing activation vectors for efficient post-training [138]. Additionally, utilizing model-native activation bases allows for both data selection and inference-time steering, grounding skill characterization in the model's own representations rather than external ontologies [146].

Evaluation challenges remain a significant bottleneck in assessing the efficacy of these adaptation strategies, particularly in quantifying the trade-off between plasticity and stability across diverse domains. Benchmarks have been introduced to evaluate models on streaming knowledge, revealing that current state-of-the-art systems struggle with state-tracking delays and susceptibility to distraction within streaming environments [254]. Similarly, utilizing Wikipedia snapshots to track temporal misalignment highlights the need for metrics that capture both retention and acquisition over time, verifying that factual knowledge can be updated with minimal training data [238]. The challenge is further compounded in specialized domains, where evaluating adaptation in real-world egocentric streams reveals that distribution shifts are often uncontrolled and long-tailed [123]. Moreover, empirical studies underscore the gap between robustness tuning before deployment and test-time adaptation, proposing benchmarks to measure performance under text corruption and domain shifts [287].

Contradictions and agreements exist regarding the optimal mechanism for knowledge integration, particularly concerning the efficacy of parameter-efficient methods versus feature-space adaptations. While some studies suggest that parameter-efficient fine-tuning methods are sufficient, others argue they introduce intruder dimensions that degrade pre-trained features, positing that operating directly in feature space avoids these issues and offers superior retention [135]. Additionally, the role of memory architecture is debated, with some contending that combining In-Weight Learning with In-Context Learning via modular memory is essential for scalable adaptation [182]. Conversely, other research suggests that outdated adapted weights can be recycled for upgraded models, challenging the notion that previous adapters must be discarded [255]. Finally, the efficacy of knowledge editing versus full retraining is a point of contention, with surveys delineating editing as a precise alternative to fine-tuning [17]. However, notes indicate that

many editing methods fail in autoregressive generation scenarios, necessitating a consolidation phase to align parametric knowledge with inference behavior [114]. These varying perspectives highlight the complexity of defining a unified framework for dynamic model adaptation.

Theoretical Foundations of Knowledge Storage, Localization, and Forgetting

Effective adaptation in artificial intelligence systems fundamentally relies on the precise identification of where knowledge resides within network architectures. Contemporary research increasingly posits that knowledge is not uniformly distributed but rather concentrated in specific modules, particularly Multi-Layer Perceptrons (MLPs) and attention mechanisms. Early investigations into the internal mechanics of transformers identified "knowledge neurons" within feed-forward layers, suggesting that specific factual associations are encoded in sparse subsets of parameters [141]. This localization hypothesis has driven the development of knowledge editing techniques that target these specific components to update model behavior without full retraining. However, recent empirical analyses challenge the exclusivity of MLPs as the sole repositories of factual information. Studies indicate that attention modules, especially in earlier layers, play a substantial role in knowledge storage and retrieval, necessitating editing strategies that balance updates across both MLP and attention components to avoid residual outdated knowledge [31]. Furthermore, distinctions exist between how different types of knowledge are stored; while entity-based knowledge may localize to MLP weights, relational knowledge appears significantly encoded within attention modules, highlighting the multifaceted nature of internal representation [199].

The mechanism of adaptation can be theoretically framed through the lens of Bayesian inference, where various adaptation methods are interpreted as corrections to approximate posteriors. According to this perspective, the efficiency of adaptation is inversely related to the accuracy of the initial posterior; more accurate priors require smaller corrections, thereby facilitating quicker learning [236]. This theoretical grounding supports the emergence of "steering" as a distinct adaptation paradigm. Unlike traditional fine-tuning which modifies weights, steering involves targeted interventions in activation space during inference. This approach enables local and reversible behavioral changes, positioning activation manipulation as a viable alternative to parameter updates for controlling model outputs [108]. The efficacy of such interventions is further supported by findings that specific skills can be unlearned or modified by detecting and adjusting key spaces within feed-forward layers or by applying rotation matrices to neural circuitries, demonstrating that behavioral control can be achieved through precise, lightweight interventions [281].

Despite the promise of localized updates, the phenomenon of catastrophic forgetting remains a critical barrier to continual learning. Forgetting is often characterized as a geometric conflict where the covariance structures induced by new tasks misalign with the evolving model state, leading to interference with previously acquired capabilities [261]. To mitigate this, researchers have proposed preserving the structural integrity of feature spaces rather than merely matching point-wise features. Methods such as Proxy-FDA align local neighborhood structures in the feature space to retain rich conceptual knowledge during fine-tuning, significantly reducing concept forgetting compared to standard regularization techniques [85]. Similarly, operating directly in feature space via element-wise transformations can prevent the introduction of "intruder dimensions" that overwrite pre-trained knowledge, offering a robust alternative to weight-space adaptations [135].

The tension between stability and plasticity has also prompted a re-evaluation of memory architectures in continual learning agents. A growing consensus suggests that relying solely on in-weight learning is insufficient for scalable adaptation. Instead, a modular memory framework that combines in-weight learning with in-context learning offers a more biologically plausible solution. In this architecture, a frozen core model handles generalization while distinct memory modules manage working and long-term knowledge accumulation, effectively decoupling rapid adaptation from stable knowledge retention [182]. This modular approach aligns with findings that knowledge circuits evolve in a deep-to-shallow pattern during continual pre-training, suggesting that new knowledge acquisition follows distinct structural phases that could be better managed through specialized memory components [263].

Furthermore, the dynamics of optimization itself play a crucial role in knowledge retention. The hyperparameters of optimizers like Adam define a statistical memory horizon, and treating these parameters as dynamic variables rather than fixed constants can improve robustness against forgetting during training [230]. Additionally, the choice of learning paradigm influences forgetting rates; reinforcement fine-tuning has been shown to inherently preserve prior knowledge better than supervised fine-tuning due to implicit regularization effects, suggesting that the optimization objective is as critical as the architectural design in maintaining knowledge integrity [122]. Ultimately, the field is

moving towards unified frameworks that view adaptation not merely as parameter updating but as a complex interplay of localization, geometric alignment, and modular memory management to achieve efficient and stable knowledge evolution.

Mechanistic Interpretability and Knowledge Localization

The foundational understanding of how Large Language Models (LLMs) store and retrieve factual knowledge has undergone a significant paradigm shift, moving from a singular focus on Multi-Layer Perceptron (MLP) modules to a more nuanced recognition of the distributed roles played by attention mechanisms. Early mechanistic interpretability research posited that factual associations are primarily localized within the feed-forward networks of middle layers, a hypothesis supported by causal tracing interventions that successfully updated specific facts by modifying MLP weights [86]. This "locate-then-edit" paradigm, exemplified by methods such as ROME and MEMIT, operated on the assumption that knowledge neurons within these MLP blocks act as the sole repositories for entity-specific information [141]. However, subsequent investigations have challenged this monolithic view, revealing that such an exclusive focus on MLPs overlooks critical components of the transformer architecture, thereby limiting the efficacy and robustness of knowledge editing strategies.

Recent empirical evidence demonstrates that attention modules play a substantial and often underestimated role in factual knowledge storage and retrieval, particularly in the earlier layers of the model. Studies conducting systematic knowledge localization experiments have found that attention modules are not merely routing mechanisms but actively participate in encoding relational knowledge [31]. This finding is corroborated by research distinguishing between entity-centric and relation-centric knowledge storage, which reveals that while entity information may be heavily concentrated in MLPs, relational knowledge is significantly encoded within attention layers [199]. Consequently, editing strategies that target only MLP weights fail to address the full scope of the knowledge triplet, leaving residual outdated information in the attention modules and resulting in incomplete updates. This limitation is particularly evident when attempting to edit relations rather than subjects, where entity-centric methods exhibit poor performance due to their inability to access the attention-based storage of relational attributes [218].

The realization that knowledge is distributed across both MLP and attention modules necessitates the development of balanced editing strategies that jointly update these components. Methods that ignore the contribution of attention mechanisms risk inducing knowledge conflicts and distorting the model's innate knowledge structure, leading to side effects such as the deterioration of general capabilities and reasoning collapse [13]. To address this, novel approaches like IntAttnEdit have been proposed, which extend the associative memory paradigm to proportionally allocate update magnitudes based on the measured contribution of both MLP and attention modules to specific knowledge storage [31]. By employing a knowledge balancing strategy, these methods achieve superior edit success rates and improved generalization compared to traditional MLP-only edits. Furthermore, cross-lingual explorations have validated the importance of attention mechanisms, showing that methods incorporating attention updates, such as MEMAT, deliver remarkable improvements in magnitude metrics and portability across languages not included in the training data [193].

Despite these advancements, the complexity of knowledge localization introduces significant challenges regarding the stability and scalability of editing operations. The interplay between different modules means that localized parameter edits can propagate along fragile directions in the representation space, potentially causing global interference and catastrophic forgetting, especially during sequential editing scenarios [208]. The phenomenon of "editing overfit," where models assign disproportionately high probabilities to edit targets at the expense of generalization, further highlights the risks of imprecise localization and overly aggressive parameter modification [211]. Moreover, the distinction between fine-grained neuron-level localization and broader module-level updates remains a critical area of inquiry. While fine-grained techniques like FiNE aim to enhance editing locality by targeting specific neurons within feed-forward networks, they must contend with the entangled nature of subject representations, where target and irrelevant knowledge often share the same semantic space [120].

The evolving understanding of knowledge localization also extends to the differentiation between long-term parametric memory and working memory representations. Research indicates that relying solely on direct parameter modification (long-term memory) or retrieval-based activation steering (working memory) presents an "impossible triangle" of reliability, generalization, and locality in lifelong editing settings [140]. This suggests that effective knowledge editing requires a hybrid approach that respects the distinct roles of various architectural components. For instance, steering vectors derived from activation spaces can be translated into component-level weight edits to achieve more precise behavioral control without the trade-offs associated with global activation interventions [155]. Additionally, the discovery that internal representations preserve syntactic structures similar to natural language allows for manipulation

strategies that replace entity knowledge blocks while maintaining the model's general abilities, offering an alternative to direct weight modification [9].

Ultimately, the transition from viewing MLPs as the exclusive vaults of knowledge to recognizing the collaborative function of attention modules marks a critical maturation in the field of mechanistic interpretability. This balanced perspective underscores the necessity for editing frameworks that are sensitive to the heterogeneous distribution of knowledge across the transformer architecture. Failure to account for the substantial role of attention mechanisms not only limits the success rate of individual edits but also exacerbates issues related to knowledge distortion and model instability in complex, multi-edit environments [28]. As the field progresses, future methodologies must continue to refine localization techniques to accurately identify and modify the specific interplay between attention and MLP components, ensuring that knowledge updates are both precise and robust against the inherent complexities of large-scale neural networks.

Geometric and Control-Theoretic Perspectives on Stability

The theoretical understanding of knowledge storage and the mitigation of catastrophic forgetting has evolved from heuristic regularization toward rigorous geometric and control-theoretic frameworks. This paradigm shift rests on the premise that neural network parameter spaces possess intrinsic manifold structures, where stable learning corresponds to navigating specific subspaces that preserve previously acquired mappings. A foundational contribution to this perspective formalizes the training of Neural Ordinary Differential Equations as a multi-motion planning problem, proving that the set of control functions mapping input points to target labels forms a Banach submanifold of finite codimension under nonsingular controls [47]. In this continuous-time framework, "Tuning without Forgetting" is not merely an approximation but represents a continuous deformation of the control function along the tangent space of this submanifold, providing a theoretical guarantee that end-point mappings for previously learned samples remain invariant during sequential updates. This geometric characterization suggests that forgetting can be solved by restricting parameter updates to the null space of the Jacobian associated with prior tasks, effectively isolating new learning from old knowledge within the control function space.

This concept of constraining updates to specific geometric subspaces extends beyond continuous-time systems to discrete transformer architectures and large language models, where the principle of orthogonality serves as a critical mechanism for stability. For instance, recent work applies null-space projection to model editing in medical vision transformers, constructing an orthogonal projection matrix from anchor samples to ensure that parameter updates strictly reside in the null space of previous representations [219]. This approach guarantees mathematically that targeted corrections do not perturb established diagnostic features, mirroring the invariance properties found in Neural ODEs. Similarly, other methods address the drift of functional directions in large language models by dynamically tracking historical function directions and projecting new gradients to be orthogonal to them, thereby mitigating interference between new and old tasks [298]. These approaches share a common geometric intuition: forgetting arises when update vectors possess non-zero components along the directions critical for prior tasks, and stability is achieved by enforcing orthogonality or confinement to specific manifolds.

However, the geometry of forgetting is not solely defined by orthogonality; it is also deeply influenced by the curvature of the loss landscape and the alignment of task geometries. Research posits that forgetting is a state-relative update-integration failure caused by the misalignment between the covariance geometry of new tasks and the evolving model state [261]. When the geometric structures induced by sequential updates conflict, interference occurs, suggesting that simple orthogonality is insufficient without considering the broader geometric tolerance of the activation space. This aligns with findings that identify a "Covariance Trap," where standard covariance constraints paradoxically amplify input perturbations, leading to sharp minima and instability [126]. The authors argue that joint optimization with orthogonal gradients can collapse solutions into regions with narrow stability tolerances, indicating that robust continual learning requires managing the curvature and isotropic alignment of the representation space rather than just enforcing linear independence.

Further refining the geometric perspective, recent work distinguishes between the geometry of weights and the geometry of activations, exploring how interventions in activation space can be translated into component-level weight edits. Some frameworks leverage steering vectors as diagnostic signals to perform rank-1 weight edits on specific attention heads and MLP neurons, redistributing behavioral influence without global activation shifts [155]. This approach acknowledges that the geometric structure of knowledge is heterogeneous across model components, and uniform updates fail to respect this granularity. Complementing this, other studies argue that adhering to Euclidean metrics for unlearning updates is sub-optimal because it ignores the geometric structure of the output probability space [173]. By embedding updates into a manifold rendered by the remaining data's geometry and utilizing Hessian information, methods can prevent effective unlearning from interfering with retained performance, highlighting the necessity of non-Euclidean geometric considerations in parameter updates.

The interplay between plasticity and stability is also governed by the spectral properties of parameter updates, offering a post-hoc geometric correction that does not require access to original training data. Some treatments view fine-tuning updates as a combination of low-rank task-aligned signals and IID-like noise residuals, where applying singular value thresholding to the weight delta matrix filters out the spectral bulk responsible for capability loss [205]. Furthermore, lifelong normalization has been shown to create a self-reinforcing stability loop, yielding parameter updates with

asymptotic orthogonality and bounded norms [44]. Despite these advances, contradictions remain regarding the universality of geometric solutions, as practical implementations in discrete, high-dimensional models often rely on approximations that may not hold under rigorous theoretical scrutiny. Challenges the sufficiency of purely geometric constraints by highlighting that strongly negative gradient similarity is the fundamental cause of forgetting, identifying specific "conflicting neurons" that induce interference even when layer-level orthogonality is maintained [117]. Additionally, warnings exist that discrete tokenization imposes an intrinsic cost on preserving continuous manifolds, creating a bottleneck where finer quantization can worsen geometric fidelity [241]. This indicates that the geometric continuity assumed in control-theoretic models may be fundamentally disrupted in discrete token-based architectures, necessitating hybrid approaches that account for both continuous deformation and discrete structural constraints.

Parameter-Efficient Continual Learning Mechanisms and Adapter Architectures

The rapid expansion of Large Language Models (LLMs) and foundation models has necessitated a paradigm shift from full fine-tuning to Parameter-Efficient Tuning (PET) methods, which aim to adapt massive models to downstream tasks while mitigating computational overhead and catastrophic forgetting. Traditional full fine-tuning often leads to "concept forgetting," where a model loses its broad pre-trained capabilities when specialized for a specific task [33]. To address this, researchers have developed architectures that update only sparse subsets of parameters or introduce lightweight adapters. A primary challenge in this domain is the trade-off between plasticity for new tasks and stability for old knowledge. Theoretical frameworks suggest that larger pre-trained models inherently reduce the sample complexity of downstream learning, a phenomenon that PET methods like adapters and low-rank adaptation leverage to provide provable improvements in efficiency [225]. However, standard PET approaches still face significant hurdles in resource-constrained environments and closed-source API settings, prompting innovations such as Easy Adaptation, which utilizes specific small models to complement large models without accessing their internal parameters [229].

A critical mechanism for preventing forgetting within adapter architectures involves managing the geometric relationship between parameter updates for new and old tasks. Catastrophic forgetting is often driven by the drift of functional directions during fine-tuning. Dynamic Orthogonal Continual (DOC) fine-tuning addresses this by tracking these directions and adjusting gradients to remain orthogonal to historical function spaces, thereby minimizing interference [298]. Similarly, the concept of "geometry conflict" posits that forgetting arises when the covariance geometries of sequential tasks misalign with the evolving model state; methods like GeometryConflict Wasserstein Merging (GCWM) utilize this insight to gate updates without requiring replay data [261]. Further refining the update process, Parameter Importance Estimation-based Continual Enhancement (PIECE) selectively updates a mere 0.1% of core parameters based on Fisher Information and second-order normalization, effectively preserving general reasoning abilities while acquiring domain-specific knowledge [2]. This aligns with findings that localized updates to sparse parameter sets can maintain transferability better than uniform updates [18].

Beyond gradient manipulation, architectural innovations in adapter design have proven essential for continual adaptation. Low-Rank Adaptation (LoRA) remains a cornerstone, yet its application in sequential tasks requires careful management to prevent mutual interference between adapters. Research indicates that merging orthogonally initialized adapters or updating only relevant task-specific weights can significantly mitigate forgetting in diffusion models and LLMs [184]. In scenarios where model evolution occurs frequently, such as with periodically updated LLMs, training-free frameworks like PORTLLM create portable model patches that allow for seamless personalization of evolved models with minimal GPU memory usage [282]. Furthermore, the PLATE method exploits geometric redundancy in pre-trained networks to construct protected update subspaces, offering explicit control over the plasticity-retention trade-off without access to past task data [60].

The interplay between different learning paradigms also influences the efficacy of PET methods. Comparative analyses reveal that Reinforcement Fine-Tuning (RFT) inherently preserves prior knowledge better than Supervised Fine-Tuning (SFT) during continual post-training, attributed to the implicit regularization of RFT and its mode-seeking nature [122]. Conversely, standard fine-tuning can be stabilized through loss-adaptive learning rates, such as the FINCH scheduler, which reduces learning rates on high-loss batches to bound per-step forgetting [178]. In vision domains, Vector-based Feature Space Adaptation (VeFA) operates directly in feature space to avoid introducing "intruder dimensions" that cause overspecialization, ensuring fine-tuned weights remain within the column space of pre-trained matrices [135]. Similarly, Proxy-FDA aligns local feature neighborhood structures using dynamic proxies to preserve rich conceptual knowledge during fine-tuning, outperforming point-wise regularization methods [85].

Despite these advances, limitations persist regarding the scalability of editing and adaptation. Sequential model editing often leads to a gradual decline in editability followed by catastrophic failure, suggesting that current parameter-modifying methods struggle with long-term accumulation of knowledge [99]. To counter this, some approaches advocate for modular memory architectures that combine In-Weight Learning with In-Context Learning, treating memory as a distinct module rather than solely updating model weights [182]. Additionally, methods like Recyclable Tuning propose reusing outdated adapted weights when models are upgraded, leveraging mode connectivity

to improve convergence for continual pre-training [255]. Theoretical investigations into gradient similarity further elucidate that conflicting neurons are the primary drivers of forgetting, leading to methods like Collaborative Neural Learning that freeze conflicting units while updating collaborative ones to theoretically eliminate forgetting under specific conditions [117]. Ultimately, the field is moving towards hybrid strategies that combine sparse parameter updates, geometric constraints, and modular architectures to achieve robust, scalable continual learning.

Selective Parameter Updates and Low-Rank Adaptation

The paradigm of continual learning in foundation models has shifted decisively from full fine-tuning toward parameter-efficient mechanisms that selectively update model weights to mitigate catastrophic forgetting. The central premise driving this transition is that only a sparse subset of parameters is critical for acquiring new domain knowledge, while the vast majority of pre-trained weights must remain static to preserve general reasoning capabilities. This philosophy is exemplified by PIECE, which utilizes Fisher Information and second-order normalization to identify and update merely 0.1% of core parameters, thereby achieving state-of-the-art performance without accessing prior training data or increasing parameter counts [2]. By localizing updates to the most relevant parameters, such methods effectively decouple the acquisition of downstream task knowledge from the preservation of foundational abilities, addressing the stability-plasticity dilemma inherent in sequential learning. Complementing this importance-driven approach, theoretical frameworks based on gradient similarity have identified specific conflicting neurons that induce forgetting, suggesting that freezing these units while updating only collaborative neurons can theoretically eliminate catastrophic forgetting under specific conditions [117].

Low-Rank Adaptation (LoRA) has emerged as a dominant architectural strategy within this selective update framework, offering a mechanism to inject trainable rank decomposition matrices into frozen layers. Research indicates that the rank of these adapters plays a pivotal role in balancing forgetting and plasticity; empirical analyses suggest that the choice of rank significantly impacts the retention of pre-training knowledge and the model's ability to adapt to subsequent tasks [24]. In medical imaging applications, combining frozen foundation backbones with task-specific LoRA modules has been shown to eliminate catastrophic forgetting entirely by design, as the backbone remains untouched while adapters handle task-specific nuances [34]. However, naive sequential training of multiple LoRA adapters can still lead to mutual interference. To mitigate this, strategies such as orthogonal initialization and selective merging of adapters have been proposed to ensure that new task updates do not overwrite the functional directions established by previous tasks [184]. Furthermore, the concept of incremental rank updates extends this logic by representing network weights as linear combinations of rank-1 matrices, allowing the model to reuse factors from previous tasks while adding minimal new parameters for current objectives, effectively achieving zero forgetting without replay buffers [105].

Beyond simple sparsity or low-rank constraints, recent advancements emphasize the geometric properties of parameter updates to control forgetting. The theory of Geometry Conflict posits that forgetting arises when the covariance geometry of a new task's update misaligns with the evolving model state; consequently, methods like GeometryConflict Wasserstein Merging utilize geometry-aware corrections to gate updates and prevent interference [261]. Complementing this, Dynamic Orthogonal Continual fine-tuning tracks the drift of functional directions and explicitly adjusts new task gradients to be orthogonal to historical directions, thereby reducing interference in long-term learning scenarios [298]. Similarly, PLATE exploits geometric redundancy in pre-trained networks to construct protected update subspaces, restricting plasticity to redundant neurons while freezing dominant feature directions to guarantee worst-case retention [60]. These geometric approaches suggest that the direction of the update is as critical as the magnitude or the number of parameters involved.

The efficacy of selective updates is further enhanced by adaptive learning strategies that modulate the update process based on loss landscapes and data characteristics. For instance, FINCH introduces a loss-adaptive learning rate schedule that reduces step sizes on high-loss batches, which are identified as primary drivers of forgetting, thereby bounding per-step forgetting without altering the optimization objective [178]. In the context of reinforcement learning, the use of on-policy data has been identified as a natural regularizer that mitigates forgetting more effectively than supervised fine-tuning, suggesting that the nature of the data distribution interacting with selective parameters is a key variable [143]. This aligns with findings that selectively updating only a sparse set of specialized parameters can improve new task accuracy by up to 7% while maintaining pre-training knowledge with negligible degradation [18].

While many methods focus on preventing interference, some frameworks explore the potential for backward knowledge transfer, where updating the model for new tasks inadvertently improves performance on old tasks. CUBER achieves this by characterizing task correlations and selectively modifying old task models when learning new ones, demonstrating that judicious parameter updates can yield positive backward transfer without data replay [286]. Conversely, gating mechanisms like STABLE evaluate candidate edits against stability budgets, measured via exact

match drop or KL divergence, and rescale or reject LoRA updates that exceed thresholds, providing a principled method for continual self-editing [74]. Collectively, these diverse approaches underscore that the future of continual learning in foundation models lies not in updating all parameters equally, but in the precise, geometry-aware, and importance-driven selection of which parameters to modify, ensuring that models remain both adaptable and stable over time.

Gated Mechanisms and Geometry-Aware Adapters

The challenge of continual adaptation in large language models necessitates mechanisms that can integrate new knowledge without compromising previously acquired capabilities, a phenomenon often referred to as catastrophic forgetting. Within the broader domain of parameter-efficient continual learning, recent advancements have shifted focus toward gated mechanisms and geometry-aware constructions that offer robust alternatives to standard regularization techniques. The STABLE framework exemplifies this shift by introducing a gated continual self-editing approach that constrains forgetting during sequential updates through Low Rank Adaptation. Unlike traditional methods that apply updates indiscriminately, STABLE evaluates each candidate LoRA update against a predefined stability budget. This evaluation utilizes metrics such as Exact Match drop, bits increase, and KL divergence to quantify the potential threat to the model's existing knowledge distribution. If an update exceeds the threshold, the framework employs a clipping procedure to rescale the update or rejects it entirely, thereby preserving the model's reliability while maintaining adaptability [74].

Complementing these gating strategies, geometry-aware approaches leverage the structural properties of the parameter space to differentiate between beneficial updates and those that induce interference. GeoEdit proposes a novel framework that utilizes the geometric relationships of parameter updates derived from fine-tuning to distinguish neurons associated with new knowledge from those critical for general knowledge preservation [163]. By employing a direction-aware knowledge identification method, GeoEdit avoids updating neurons with directions approximately orthogonal to existing knowledge, thus safeguarding the model's generalization ability. For neurons where updates are necessary, the method integrates old and new knowledge for aligned directions and applies a "forget-then-learn" strategy for opposing directions, further enhanced by an importance-guided task vector fusion technique [163]. This geometric perspective is reinforced by findings in Geometry Conflict, which posits that forgetting arises when the covariance geometries induced by new tasks misalign with the geometry of the evolving model state [261]. Consequently, methods like GeometryConflict Wasserstein Merging utilize geometry conflict as a control signal to gate geometry-aware corrections, ensuring that sequential updates remain compatible with the model's current state [261].

The interplay between gating and geometric constraints is also evident in methods addressing the generality-locality trade-off, particularly in multi-modal contexts. BalancEdit introduces a mechanism that dynamically balances these competing objectives by generating positive and negative samples to determine the influence scope of each fact [267]. This approach visualizes the model's semantic similarity as dynamic spheres in latent space, where the radius is determined by the distance between samples, allowing for immediate edits while preserving correct behaviors for unrelated inputs [267]. Similarly, BetaEdit refines the null-space paradigm by integrating history-aware updates to control knowledge leakage, effectively constraining updates to preserve the model's original behavior while mitigating performance degradation during sequential editing [198]. These methods collectively suggest that successful continual learning requires not just parameter efficiency, but a sophisticated understanding of the geometric and statistical boundaries within which edits can safely occur.

Further enhancing stability, Lifelong Normalization has been identified as a core strategy shared by editors resilient over long horizons, normalizing value gradients using running statistics to create a self-reinforcing stability loop [44]. Theoretical analysis indicates that when combined with ridge-regularized regression, this normalization yields parameter updates with asymptotic orthogonality and bounded norms, directly mitigating systemic collapse [44]. Building on this, STABLEEDIT strengthens the stability loop via an explicit warm-up stage and full whitening, improving long-horizon stability with minimal overhead [44]. In parallel, Multiplicative Orthogonal Sequential Editing addresses numerical instability by multiplying the original parameter matrix by an orthogonal matrix rather than adding an update matrix, ensuring that numerical stability indicators such as condition number and norm remain unchanged [36]. This multiplicative paradigm effectively limits deviations in the edited parameter matrix, retaining a high percentage of general abilities across downstream tasks compared to additive frameworks [36].

The integration of these geometric and gating principles extends to adapter architectures designed for specific modalities and tasks. For instance, in the realm of diffusion models, continual customization is achieved by merging orthogonally initialized adapters or updating only relevant task-specific weights to mitigate mutual interference [184]. Similarly, Dynamic Orthogonal Continual fine-tuning tracks the drift of functional directions and adjusts gradients of new task parameters to be orthogonal to tracked historical function directions, effectively reducing catastrophic

forgetting in large language models [298]. These approaches underscore a consensus in the literature: preserving the integrity of the pre-trained representation space through geometric constraints and gated updates is paramount for scalable and reliable continual learning. By moving beyond simple regularization to embrace geometry-aware adapters and stability budgets, these frameworks provide a principled method for large language models to integrate new knowledge while maintaining the reliability and robustness required for real-world deployment.

Kernel-Based and Latent Space Approaches

Within the broader landscape of Parameter-Efficient Continual Learning mechanisms, a distinct paradigm has emerged that shifts the locus of adaptation from weight space to feature and latent spaces. Rather than updating the parameters of the foundation model backbone or even lightweight adapters, these approaches leverage the rich, frozen representations of pre-trained models to achieve class separation and task integration. A primary challenge in Class-Incremental Learning involves mitigating catastrophic forgetting while ensuring inter-task class separation. Recent advancements suggest that directly utilizing frozen features can be suboptimal without further transformation. To address this, Kernel Linear Discriminant Analysis introduces a method that leverages the Radial Basis Function kernel and Random Fourier Features to enhance feature representations from frozen foundation models [11]. By computing class means and updating a shared covariance matrix based on these kernelized features, this approach achieves accuracy comparable to joint training upper bounds without relying on replay data, effectively resolving both forgetting and separation issues [11]. This aligns with findings that preserving the original feature space of foundation models is critical, as standard fine-tuning often induces concept forgetting, where the model loses its ability to recognize concepts unrelated to the downstream task [33].

The efficacy of operating in latent space extends beyond kernel methods to include prototype-based and statistical feature combinations. In medical imaging, specifically dermatological classification, research demonstrates that freezing the foundation model backbone and training only a lightweight MLP or employing a zero-training nearest-mean classifier on prototypes derived from embeddings can outperform complex regularization and replay-based methods [77]. This supports the notion that the representational power of frozen foundation models is sufficient for continual learning if the classification head is appropriately designed. Similarly, in time-series classification, combining deep temporal embeddings from frozen foundation models with statistical features in a dual-stream pipeline allows for competitive accuracy and low forgetting rates without the need for extensive parameter updates [57]. These approaches underscore a consensus that the latent space of foundation models contains robust, transferable knowledge that, when accessed via non-parametric or lightweight parametric heads, can sustain continual learning performance [250].

However, the stability of latent space approaches is not universal and depends heavily on the nature of the data and the specific adaptation strategy. While some studies advocate for completely frozen backbones to prevent interference, others propose dynamic adjustments within the latent space to handle distribution shifts. For instance, in EEG foundation models, the EvoBrain framework utilizes Neuro-Spectral Task Normalization to align incoming tasks with historical statistics in the latent space, facilitating adaptation across heterogeneous Brain-Computer Interface tasks without full retraining [61]. Conversely, investigations into time series foundation models reveal that sequential fine-tuning can still induce significant degradation on previously learned tasks, highlighting a persistent stability-plasticity dilemma even in powerful latent representations [5]. This contradiction suggests that while frozen features are powerful, the mechanism for integrating new class information into the latent space, whether through kernel transformations, prototype updates, or spectral normalization, must be carefully tailored to the domain's temporal and structural characteristics.

Furthermore, the theoretical underpinnings of these methods often draw from the concept that neural networks can be interpreted as ensembles of fixed functions in the lazy regime, where posterior updates mimic scaled stochastic gradient descent [79]. This perspective validates the use of fixed encoders with evolving classifiers, as seen in latent replay strategies where replay is performed in the latent space rather than the raw data space [25]. Such latent replay has been shown to offer a favorable compute-accuracy trade-off, particularly when the pre-trained encoder possesses high representational similarity to the downstream tasks. Nevertheless, challenges remain regarding the geometry of these latent spaces. Research indicates that forgetting can be viewed as a state-relative update-integration failure arising from conflicts in the covariance geometry induced by sequential tasks [261]. Consequently, advanced latent space approaches are beginning to incorporate geometric constraints, such as null-space projections, to ensure that updates for new classes do not perturb the representations of anchor samples, thereby mathematically guaranteeing the preservation of prior knowledge [219].

The integration of kernel methods and latent space manipulation represents a significant departure from traditional weight-space fine-tuning. Methods like VeFA operate directly in feature space to avoid generating intruder dimensions

that lead to overspecialization and forgetting, applying element-wise transformations that keep fine-tuned weights within the column space of pre-trained matrices [135]. Similarly, the use of spline additive models like KASAM attempts to provide intrinsic memory retention through function approximation in a way that traditional neural networks struggle with, although it requires pseudo-rehearsal to fully mitigate forgetting [185]. Collectively, these works illustrate that the future of parameter-efficient continual learning lies not merely in reducing the number of trainable parameters, but in strategically navigating and transforming the high-dimensional latent spaces provided by foundation models to maintain a balance between plasticity for new tasks and stability for old ones.

Advanced Model Editing Techniques for Factual Updates and Unlearning

Model editing has emerged as a critical paradigm for updating Large Language Models (LLMs) without the prohibitive computational costs associated with full retraining or fine-tuning. The primary objective is to inject new factual knowledge or correct errors while preserving the model's general capabilities and unrelated knowledge. However, scaling these techniques to handle sequential or massive batches of edits reveals significant challenges, particularly regarding catastrophic forgetting and the degradation of editing efficacy over time. Research indicates that state-of-the-art methods like ROME and MEMIT, while effective for single or few-shot edits, suffer from progressive performance decline when applied sequentially. Specifically, as the number of edits increases, models enter a phase of gradual forgetting followed by an abrupt, catastrophic loss of both edited facts and downstream task performance [99]. This scalability bottleneck suggests that direct parameter modification strategies often fail to maintain the delicate balance between reliability, locality, and generalization required for lifelong adaptation.

To address the limitations of direct parameter updates, recent literature has explored architectural innovations and memory-based approaches. A prominent direction involves decoupling edited knowledge from the base model parameters to avoid interference. One study proposes a dual parametric memory scheme, separating pretrained knowledge into a main memory and edited knowledge into a side memory, governed by a router to resolve the "impossible triangle" of reliability, generalization, and locality in lifelong settings [140]. Similarly, another approach introduces a system that stores edits in an explicit memory and utilizes a counterfactual model to reason over them, achieving superior performance across diverse editing problems compared to parameter-modifying baselines [110]. Further advancing this concept, a brain-inspired architecture with distributed episodic memory enables dynamic one-shot updates and selective forgetting with significant speed improvements [104]. These memory-augmented strategies contrast with methods that achieve ultra-scalable editing through lightweight linear algebra operations and lifelong normalization, demonstrating the ability to support up to one million edits while maintaining accuracy on consumer-grade hardware [4].

The challenge of preserving historical knowledge while integrating new information has led to the formulation of Temporal Knowledge Editing. Standard editing often conflates error correction with knowledge updates, causing the model to forget valid historical facts, which necessitates frameworks that concurrently edit historical and new knowledge based on temporal context [266]. In parallel, efforts to improve the stability of sequential updates have yielded theoretical frameworks that model editing as a constrained stochastic programming problem using Lyapunov optimization to guarantee long-term knowledge preservation [148]. Additionally, research identifies lifelong normalization as a key mechanism for stability, proposing methods to strengthen this effect through explicit warm-up stages and full whitening [44].

Beyond scalability and temporal dynamics, the precision of knowledge localization and the scope of influence remain critical areas of investigation. Traditional locate-then-edit methods often struggle with the generality-locality trade-off, particularly in multi-modal contexts. Recent work addresses this by dynamically determining the influence scope of edits using positive and negative samples, ensuring that updates generalize appropriately without affecting unrelated inputs [267]. At a finer granularity, other techniques advocate for neuron-level interventions, optimizing hidden states and iteratively selecting neurons based on activation values to mitigate forgetting [14]. Complementing these, some approaches manipulate internal entity representations via a hypernetwork, enabling massive simultaneous edits by replacing knowledge blocks without altering model weights [9].

The application of model editing extends to unlearning, where the goal is to remove unwanted information rather than replace it. Investigations into adapting editing algorithms for unlearning tasks find that they can exceed baseline unlearning methods in quality but still struggle with defining the precise scope of removal [62]. Proposed frameworks for this purpose utilize a three-component objective function that combines reverse gradients for unlearning, gradient descent for updates, and KL divergence minimization for retention [119]. Furthermore, generalized forgetting updates have been developed for coordinated multi-layer edits in vision models, identifying critical layers and heads to ensure precise unlearning without damaging overall performance [253].

Despite these advancements, fundamental limitations persist regarding the robustness and generalization of edited models. Theoretical analyses based on the dimensional collapse hypothesis argue that localized parameter edits can

induce global interference and reasoning collapse, suggesting that retrieval-based baselines may outperform parameter-editing methods in realistic settings [208]. This concern is echoed in studies highlighting risks of knowledge conflict and distortion, where editing logically clashing facts magnifies inconsistencies and warps the model's innate knowledge structure [28]. To mitigate overfitting and improve generalization, researchers have identified "Editing Overfit" as a prevalent issue and proposed strategies to guide models in recalling new knowledge similarly to in-context learning [211]. Moreover, findings suggest that residual single-hop knowledge undermines multi-hop reasoning, proposing an erasure mechanism to clear old pathways before injecting new knowledge [20]. These findings collectively underscore the complexity of achieving robust, scalable, and precise model editing, necessitating a shift from simple parameter tweaks to more sophisticated, theoretically grounded, and memory-aware frameworks.

Direct Parameter Modification and Null-Space Constraints

Traditional approaches to updating Large Language Models (LLMs) often rely on direct parameter modification, yet these methods frequently struggle with imprecise localization, leading to unintended alterations in unrelated knowledge. The fundamental challenge lies in distinguishing the specific neurons responsible for a particular fact from those supporting general model capabilities. Early foundational work established that factual knowledge is stored in localized, editable computations within feed-forward modules, prompting the development of "locate-then-edit" strategies [141]. However, simply identifying a layer is often insufficient; the coupling of parameters and the lack of dynamic interaction across layers can result in poor editing locality and the persistence of inaccurate facts [8]. To address imprecise localization, the Knowledge Neuronal Ensemble (KNE) method computes gradient attribution scores for each parameter at every layer, updating only the specific ensemble of neurons encoding the target knowledge. This approach mitigates the issue of frequent parameter modification caused by coupling and ensures collaborative updates among parameters, significantly improving accuracy and portability [8].

Building on the need for finer granularity, recent research argues that existing locate-then-edit methods are often sensitive only to changes in the subject entity, failing to adapt effectively to changes in relations. This limitation stems from insufficient precision in knowledge localization. The Fine-grained Neuron-level Knowledge Editing (FiNE) method addresses this by precisely identifying and modifying specific neurons within feed-forward networks, thereby enhancing editing locality without compromising overall success rates [120]. Similarly, the concept of "golden layers" suggests that different queries localize knowledge at different depths. The Layer Gradient Analysis (LGA) method utilizes gradient attribution to efficiently estimate these optimal layers, avoiding extensive trial-and-error and ensuring that edits are applied where they are most effective [37]. Furthermore, the assumption that knowledge is isolated within specific modules has been challenged; studies reveal that attention modules also play a substantial role in factual storage, particularly in earlier layers. Consequently, methods like IntAttnEdit propose a knowledge balancing strategy to jointly update both Multi-Layer Perceptron (MLP) and attention modules, preventing residual outdated knowledge from limiting editing effectiveness [31].

A critical aspect of direct parameter modification is managing the geometric constraints of the update to preserve the model's null space—the subspace of parameters unrelated to the target knowledge. Geometric Knowledge Editing (GeoEdit) utilizes the geometric relationships of parameter updates from fine-tuning to differentiate between neurons associated with new knowledge updates and those related to general knowledge perturbations. By employing a direction-aware knowledge identification method, GeoEdit avoids updating neurons with directions approximately orthogonal to existing knowledge, thus preserving the model's generalization ability. For aligned directions, it integrates old and new knowledge, while applying a "forget-then-learn" strategy for opposite directions [163]. This geometric perspective is further supported by findings that standard covariance constraints can act as a "Covariance Trap," amplifying input perturbations and causing generalization collapse. The Robust Same-subject Editing (RoSE) method counters this by employing isotropic geometric alignment to minimize representational deviation, ensuring that inner activation drifts do not exceed the model's geometric tolerance [126].

The challenge of maintaining stability becomes even more pronounced in lifelong or sequential editing scenarios, where accumulated errors can lead to catastrophic forgetting. Neuron-Specific Masked Knowledge Editing (NMKE) combines neuron-level attribution with dynamic sparse masking, utilizing an entropy-guided mask to locate relevant neurons while distinguishing between knowledge-general and knowledge-specific neurons. This strategy enables precise editing with fewer parameter modifications, maintaining high success rates over thousands of sequential edits [35]. Similarly, Neuron-level Sequential Editing (NSE) optimizes target layer hidden states using original weights to prevent model failure and iteratively selects neurons based on activation values to mitigate forgetting [14]. However, theoretical analyses warn of a "dimensional Collapse Hypothesis," where localized edits propagate along fragile directions in the representation space, inducing global interference and reasoning collapse. Empirical evidence suggests that under realistic conditions, parameter-based editing methods may consistently damage core LLM capabilities compared to retrieval-based baselines, highlighting the inherent risks of direct modification [208].

To further refine the scope of edits and prevent overfitting, some approaches focus on disentangling knowledge representations. The DiKE method decomposes subject representations into target-related and unrelated components, updating only the former to preserve fine-grained irrelevant knowledge [247]. Additionally, the phenomenon of

"Editing Overfit," where models assign disproportionately high probabilities to edit targets, hinders generalization in complex multi-hop reasoning tasks. The Learn the Inference (LTI) strategy introduces multi-stage inference constraints to guide edited models in recalling new knowledge similarly to in-context learning, effectively mitigating this overfitting [211]. While methods like Rank-One Model Editing (ROME) have demonstrated effectiveness in modifying feed-forward weights to update specific associations [86], the trade-off between generality and locality remains a central concern. BalancEdit addresses this in multi-modal settings by dynamically determining the influence scope of each fact, generating positive and negative samples to balance the edit's reach without modifying underlying weights [267]. Ultimately, while direct parameter modification offers a computationally efficient alternative to retraining, the integration of null-space constraints, geometric alignment, and fine-grained neuron selection is essential to ensure that edits are precise, stable, and non-destructive to the model's broader knowledge base.

Scalability, Massive Editing, and Unlearning

Scaling knowledge editing to encompass thousands or even millions of factual updates remains a formidable obstacle in the deployment of Large Language Models (LLMs), as traditional locate-then-edit methods often degrade significantly when subjected to massive sequential or batch operations. A primary challenge in this domain is the phenomenon of catastrophic forgetting, where models lose previously edited facts or general capabilities as the number of edits increases, often following a trajectory of gradual decline followed by abrupt collapse [99]. This limitation necessitates the development of architectures and algorithms specifically designed for high-volume scalability without compromising the model's foundational reasoning abilities or numerical stability.

To address the scalability bottleneck, recent advancements have shifted towards dynamic weight generation and hypernetwork-based solutions that decouple the storage of edits from the base model parameters. The MeG framework introduces a novel approach by attaching dynamic weight neurons to specific layers and utilizing diffusion models to conditionally generate weights based on input queries, allowing for massive editing with a single dynamic neuron per layer [16]. This method significantly improves reliability, generality, and particularly locality metrics compared to existing techniques by avoiding the direct accumulation of parameter shifts that often leads to interference. Similarly, the Latent Knowledge Scalpel manipulates internal entity representations via a lightweight hypernetwork, demonstrating the capacity to execute up to 10,000 simultaneous edits while preserving the model's general syntactic and semantic structures [9]. These methods contrast with earlier hypernetwork approaches that struggled with statistical significance and cancellation effects when aggregating parameter shifts for synchronous editing, a problem addressed by formulating parameter shift aggregation as a least squares problem to enable arbitrary batch sizes [38].

Another critical avenue for achieving massive editing involves optimizing memory structures and addressing the root causes of performance degradation, such as embedding collisions. The NAMET method posits that the failure of existing approaches in massive editing scenarios is largely due to embedding collisions among knowledge items, which undermine reliability at scale [293]. By introducing noise during memory extraction, this approach enhances robustness in context-rich settings, outperforming baseline methods when editing thousands of facts. Complementing this, the SERAC framework stores edits in an explicit external memory, employing a retrieval-augmented counterfactual model to modulate predictions, offering higher expressiveness and capacity for complex editing problems where purely parametric methods fail [110]. Furthermore, unifying theoretical frameworks have emerged, such as EMMET, which generalizes previous algorithms under a single preservation-memorization objective, enabling batched editing of up to 10,000 facts with performance comparable to state-of-the-art methods [245].

The trade-off between generality and locality becomes increasingly pronounced as the scale of editing expands, particularly in multi-modal contexts where the influence scope of facts varies dynamically. The BalancEdit method explicitly addresses this by generating positive and negative samples for each fact to determine its influence scope, utilizing a discrete codebook to balance these competing metrics without altering underlying weights [267]. For lifelong editing scenarios where edits occur continuously over time, methods like LEMoE and ULTRAEDIT have been proposed to mitigate order sensitivity and catastrophic forgetting. LEMoE leverages a Mixture of Experts adaptor with specialized routing to enhance routing consistency, while ULTRAEDIT achieves training-free, ultra-scalable editing through lightweight linear algebra operations, supporting up to one million edits with minimal VRAM consumption [4]. Theoretical analyses suggest that parameter-based editing can induce global interference and reasoning collapse if not carefully constrained, highlighting the need for methods that preserve core capabilities during both editing and unlearning processes [208].

Parallel to the challenge of adding knowledge is the critical task of unlearning, or removing sensitive, incorrect, or obsolete information, which requires the complete erasure of information traces rather than simple redirection. Investigations reveal that editing algorithms can be adapted for unlearning by designing specific targets, though they often struggle to define the precise scope of removal without damaging overall performance [62]. To enhance precision in unlearning, particularly for vision transformers and diffusion models, the RAZOR framework employs a ratio-aware optimization strategy that identifies and edits only the most critical layers and attention heads contributing to the target data [253]. This ensures efficient forgetting while retaining useful knowledge across diverse modalities, addressing the difficulty of encapsulating the scope of what is to be unlearned.

Despite these advancements, significant hurdles remain regarding the stability of sequential edits and the preservation of numerical stability in parameter matrices. The MOSE method proposes a multiplicative orthogonal editing paradigm to maintain the condition number and norm of parameter matrices, thereby preventing the degradation of general abilities observed in additive frameworks during sequential editing [36]. Moreover, the concept of "Lifelong Normalization" has been identified as a key mechanism in stabilizing long-horizon editing, where normalizing value gradients creates a self-reinforcing stability loop that mitigates systemic collapse [44]. As the field progresses, the integration of knowledge graphs to track associated changes and the development of visual analytics systems to guide layer selection represent promising directions for achieving truly scalable, robust, and safe model editing and unlearning [128].

Handling Unstructured, Multi-Hop, and Commonsense Knowledge

The landscape of knowledge editing has historically been dominated by approaches targeting structured, fact-based triplets, often overlooking the complexity inherent in real-world information which is frequently unstructured, multi-hop, or grounded in commonsense reasoning. Early foundational methodologies established the feasibility of locating and modifying specific feed-forward network weights to update factual associations, yet these techniques frequently encounter significant difficulties when applied to long-form content, noisy data, or intricate reasoning chains. The limitations of local layer key-value storage and term-driven optimization become particularly pronounced when dealing with unstructured data, as these methods often fail to capture the comprehensive and contextual nature of such knowledge. To address these deficiencies, recent research proposes a paradigm shift toward non-local block key-value storage and cause-driven optimization, allowing for the direct editing of the last token while preserving the surrounding context, thereby extending editing capabilities beyond structured facts to unstructured domains. Similarly, new autoregressive editing paradigms tackle the efficacy barrier associated with single-token hidden state edits by decomposing long-form knowledge into sequential chunks and iteratively editing key tokens, a strategy grounded in the Chain Rule of Mutual Information that supports diverse formats including poetry, code, and mathematical derivations.

The challenge of maintaining causal dependencies in unstructured editing is further explored by methods introducing Matryoshka-style objectives to prevent the disruption of relationships between early memory updates and later output tokens, a common failure mode in window-based autoregressive approaches. Beyond mere format, the nature of the knowledge itself poses significant hurdles, particularly regarding multi-hop reasoning where the answer is an entailed consequence of an edit rather than the edit target itself. Benchmarks designed to assess multi-hop questions have revealed that while models can successfully recall edited facts, they often fail catastrophically on complex queries, indicating a lack of ripple effects in the model's belief system. This deficiency is frequently attributed to residual single-hop knowledge that causes models to revert to original answers during complex reasoning processes. Methods addressing this issue incorporate knowledge erasure mechanisms alongside new knowledge injection, optimizing recall vectors to ensure that residual information does not undermine multi-hop performance. Furthermore, the integration of external structures, such as knowledge graphs, attempts to track associated knowledge changes, ensuring that edits reflect not only the target fact but also its relational consequences, thereby improving generalization in multi-hop scenarios.

Commonsense knowledge presents a distinct set of challenges due to its probabilistic nature and the existence of multiple correct answers, contrasting sharply with the deterministic single-answer format of encyclopedic facts. Research indicates that directly applying standard editing algorithms designed for factual recall to commonsense domains yields sub-par results. Consequently, specialized adaptations have been developed which vary edit tokens and refine layer selection strategies to better handle the nuances of commonsense judgments, demonstrating superior performance on datasets requiring probabilistic reasoning. The difficulty in editing commonsense and multi-hop knowledge is often linked to how different types of knowledge are localized within the transformer architecture. While traditional views posit that factual knowledge resides primarily in Multi-Layer Perceptron modules, recent investigations suggest that relational knowledge and commonsense reasoning are also significantly encoded in attention layers, particularly in earlier layers. This finding challenges the entity-centric focus of many editing methods and supports the development of relation-centric approaches that jointly update both MLP and attention modules to avoid leaving residual outdated knowledge that hampers reasoning.

The interplay between locality and generality becomes increasingly critical when handling unstructured and multi-hop edits, as methods enforcing strict locality may fail to propagate necessary changes across reasoning chains, while those prioritizing generality risk over-editing and hallucination. Recent approaches in the multi-modal domain address this by dynamically balancing the generality-locality trade-off through discrete codebooks of edits, determining the influence scope of each fact based on positive and negative samples. In the realm of sequential and lifelong editing, which is essential for accumulating unstructured knowledge over time, issues of catastrophic forgetting and performance degradation are prevalent, prompting proposals for dual memory schemes and normalization strategies to maintain stability across thousands of edits. Despite these advancements, the phenomenon of editing overfit, where models assign disproportionately high probabilities to edit targets at the expense of generalization, remains a concern, prompting strategies to guide models in recalling new knowledge more naturally through multi-stage inference.

constraints. The synthesis of these diverse approaches highlights a clear trajectory toward more holistic editing frameworks that respect the interconnected, unstructured, and reasoning-heavy nature of human knowledge.